Set :12

QUESTION 1 : A company needs a homepage layout. Design using Bootstrap grid with header, content and footer.

AIM :

To design and develope a responsive web page layout using Bootstrap grid system consisting of header, content and footer.

PROCEDURE / ALgorithm

The development of this page follows these logical steps.

1. Link the Styles : The code starts by connecting to the Bootstrap libary online. This gives us access to pre-made designs, so we don't write the all CSS from stratch.

2. Set the Page height : We tell the body of the page to fill the entire height of the screen. This ensures the footer stays at the very bottom even if there isn't much text.

3. Create a Dark header : A black bar is added to the top of the page with a word " AB company " Centered in white text.

4. Build the Grid layout : We create a main container in the middle of the page inside this, we split the space into two sections.

- Main part : Takes up most of the space on the left.
- Side bar : A smaller box on the right with rounded corners.

5. Make it Responsive: We use "Column" setting (col-md-8 and col-md-4) so that the two sections sit side by side on a laptop but stack on top of each other on a phone screen.

6. Add the footer : A final dark bar is placed at the bottom to show the copyright information

SOURCE CODE

```
<!DOCTYPE html>
<html lang ="en">
<head>
    <meta charset ="UTF-8">
    <meta name ="viewport" content ="width = device-
      width , initial-scale = 1.0">
    <link href = "https://cdn.jsdelivr.net/npm/bootstrap@
      5.3.0/dist/css/bootstrap.main.css" rel ="Stylesheet">
<head>
<body>
<body class ="d-flex flex-column vh-100 text-center">
<header class ="bg-dark text-white p-3">
<header class = "bg-dark text-white p-3">
  <h1> AB company </h1>
 </header>
<main class = "Container my-auto py-5 text-start">
<div class ="row">
<div class = "col-md-8">
<h2> Main Content </h2>
<p> Simplified layout using Bootstrap grid. </p>
</div>
```

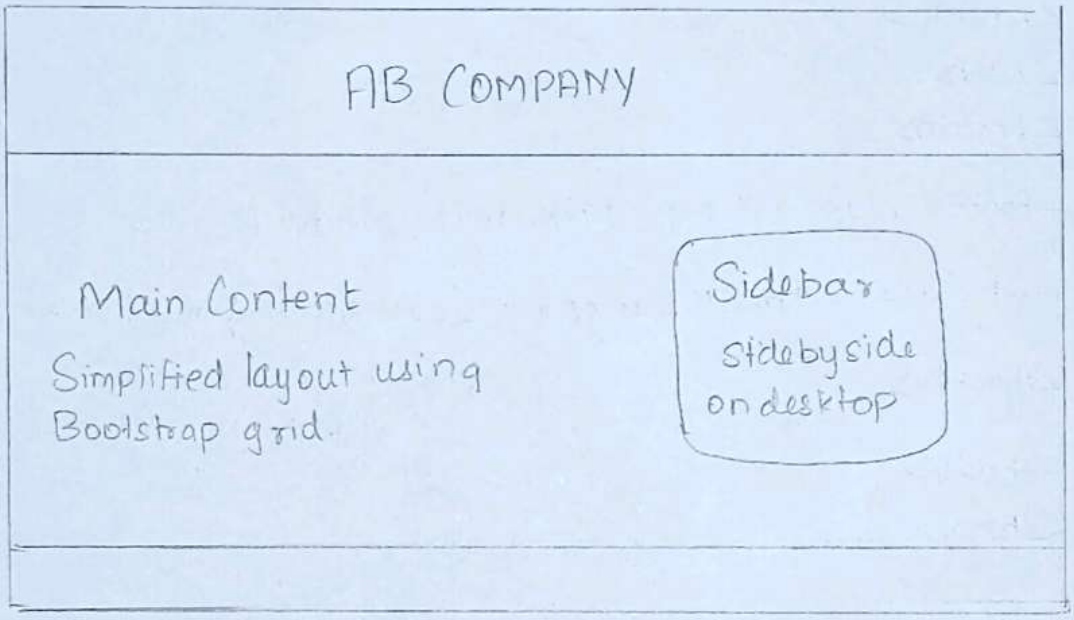
```
<aside class="col-md-4 bg-light border p-4 rounded">
  <h3> Sidebar <h3>

 <p> Stack on mobile, Side by side on mobile. </p>
</aside>
</div>
</main>

<footer class = " bg-dark text-white p-3 ">

  <p class = "m-0" > &copy ; 2026 AB company </p>

</footer>

</body>
</html>
```

RESUT :

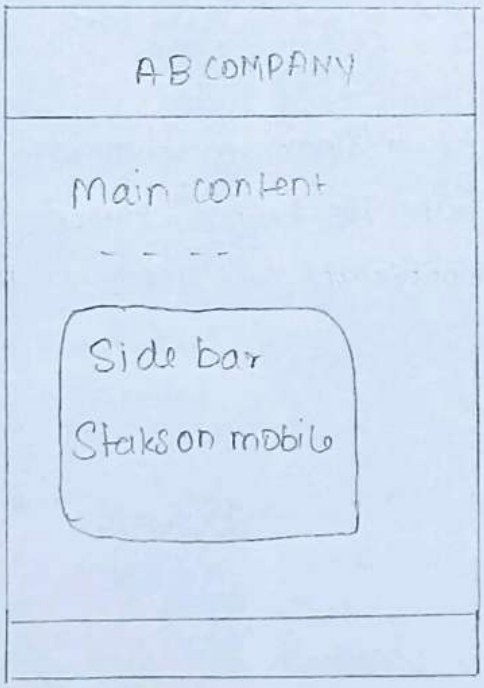
The code produces a clean ,responsive webpage that automatically adjusts its layout. Hence the output for the code using Bootstrap has been displayed.

OUTPUT.

In laptop mode.



In mobile mode.



QUESTION 2 : Write a Shell script to print number from 1 to 100.

AIM

To create a simple shell script that uses a loop to display numbers from 1 to 100. in sequence.

ALGORTTHM

The logic follows a basic "count and repeat" structure.

1. Start : initialize the script using command line "nano filename.text".
2. Initialize loop : Set a "for loop" that having index "i" Start at 1 and end at 100.
3. Process : By using "DO' it will start to count the numbers, then we will use "echo $i" to print the value of index. after finilize the loop we put "done" in the end of the code.

   - It ~~is~~ will stop after the index reach the value 100.

RESULT :

Hence we finally print the values from 1 to 100 by using shell script. It will be established in output.

SOURCE CODE :

// $ nano numlist.sh.

code.

```
for (( i=1; i<=100; i++ ))
do
echo $i
done
```

//Saved by ctrl+o and executed.

OUTPUT :

The values from 1 to 100

1
2
3
4
5
6
⋮
91
92
93
94
95
96
97
98
99
100.